NAME IT, DRAW IT!: ACTIVITY 3

HANS ANDREY R. MONREAL

CHEMICAL ENGINEERING 2A

NAME

FLIGHT

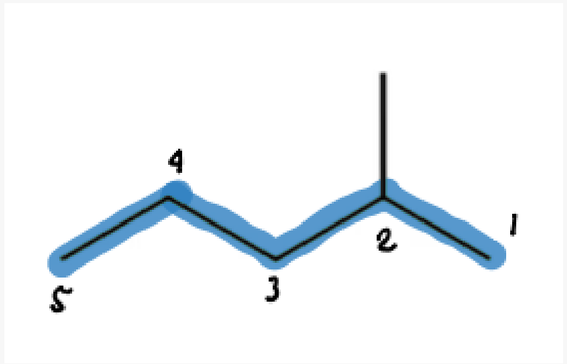
GATE

SEAT



CARD A01 - 2-methylpentane
Level: Foundation Molecular Formula: C_6H_{14}

1. *Numbered Parent*



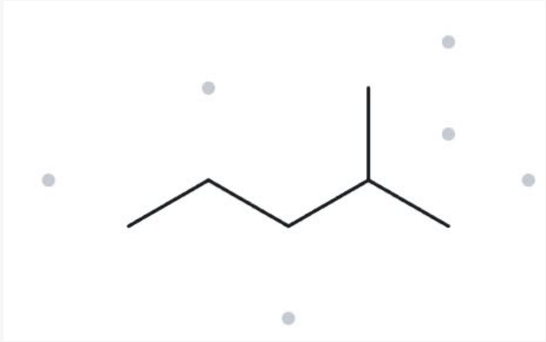
2. *Naming Trail*

- **Parent chain:** Pentane (5 carbon atoms).
- **Numbering:** Number from the end nearest the methyl branch; the branch is on carbon 2.
- **Substituent:** One methyl group at C-2. Final name: 2-methylpentane.

3. *My Name*

2-methylpentane

4. *The Alkane Check*



2-methylpentane

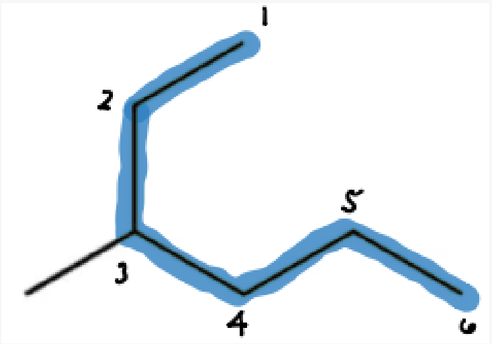
Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A02 - 3-methylhexane

Level: Foundation Molecular Formula: C_7H_{16}

1. *Numbered Parent*



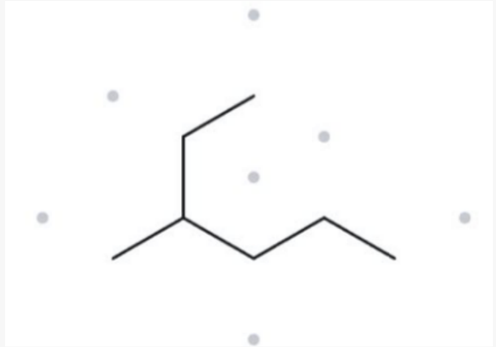
2. *Naming Trail*

- **Parent chain:** Hexane (6 carbon atoms).
- **Numbering:** Number from the end that gives the methyl branch the lower locant, C-3.
- **Substituent:** One methyl group at C-3. Final name: 3-methylhexane.

3. *My Name*

3-methylhexane

4. *The Alkane Check*



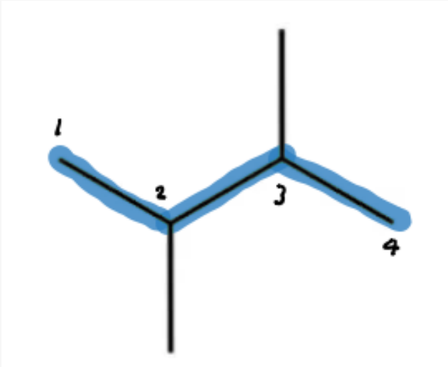
3-methylhexane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A03 - 2,3-dimethylbutane
Level: Foundation Molecular Formula: C_6H_{14}

1. *Numbered Parent*



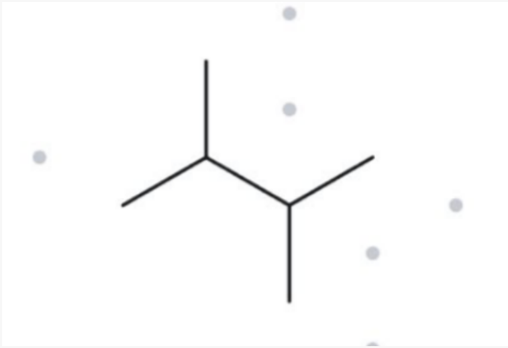
2. *Naming Trail*

- **Parent chain:** Butane (4 carbon atoms).
- **Numbering:** Number the chain to give the methyl groups locants 2 and 3.
- **Substituents:** Two methyl groups at C-2 and C-3. Final name: 2,3-dimethylbutane.

3. *My Name*

2,3-dimethylbutane

4. *The Alkane Check*



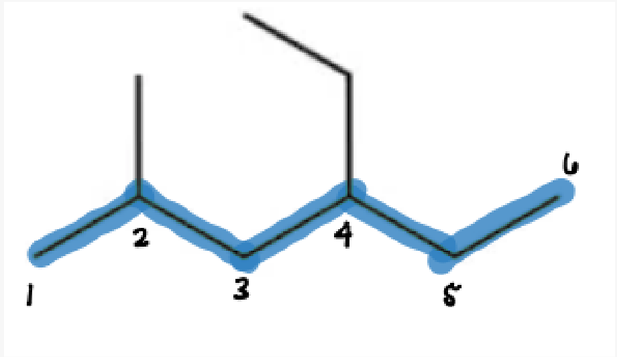
2,3-dimethylbutane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A04 - 4-ethyl-2-methylhexane
Level: Developing Molecular Formula: C_9H_{20}

1. *Numbered Parent*



2. *Naming Trail*

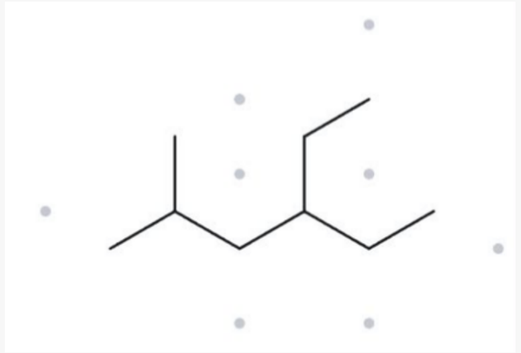
- **Parent chain:** Hexane (6 carbon atoms).
- **Numbering:** Number from the end that gives the substituents the lowest locants: 2 and 4.
- **Substituents:** Ethyl at C-4 and methyl at C-2; arrange alphabetically.

Final name: 4-ethyl-2-methylhexane.

3. *My Name*

4-ethyl-2-methylhexane

4. *The Alkane Check*



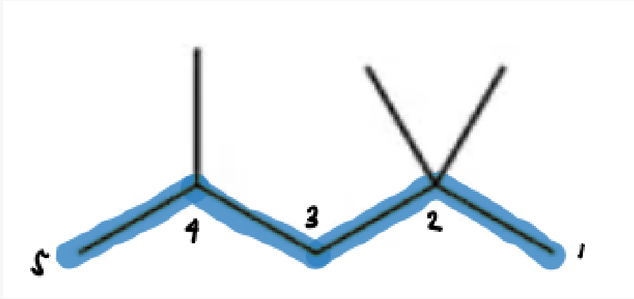
4-ethyl-2-methylhexane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A05 - 2,2,4-trimethylpentane
Level: Developing Molecular Formula: C_8H_{18}

1. *Numbered Parent*



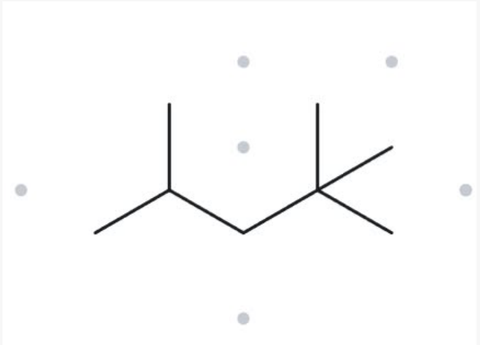
2. *Naming Trail*

- **Parent chain:** Pentane (5 carbon atoms).
- **Numbering:** Number from the end that gives the methyl groups the lowest locants: 2,2,4.
- **Substituents:** Three methyl groups at C-2, C-2, and C-4. Final name: 2,2,4-trimethylpentane.

3. *My Name*

2,2,4-trimethylpentane

4. *The Alkane Check*



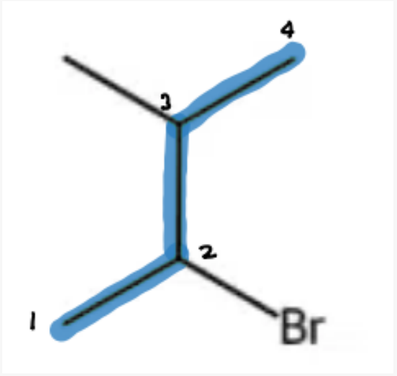
2,2,4-trimethylpentane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A06 - 2-bromo-3-methylbutane
Level: Developing Molecular Formula: $C_5H_{11}Br$

1. *Numbered Parent*



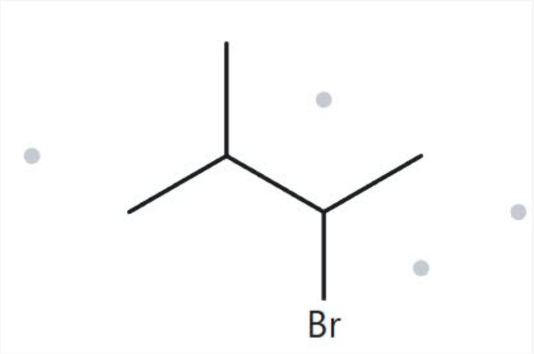
2. *Naming Trail*

- **Parent chain:** Butane (4 carbon atoms).
- **Numbering:** The two substituents have locants 2 and 3 either way; alphabetical priority gives bromo the lower number.
- **Substituents:** Bromo at C-2 and methyl at C-3. Final name: 2-bromo-3-methylbutane.

3. *My Name*

2-bromo-3-methylbutane

4. *The Alkane Check*



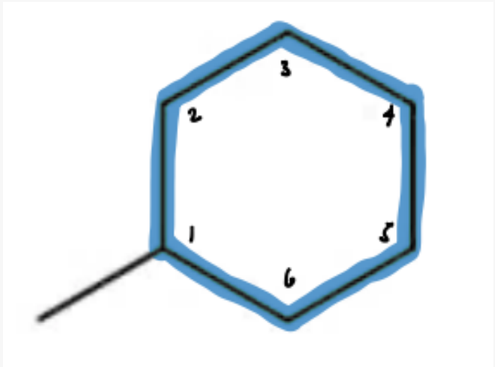
2-bromo-3-methylbutane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A07 - methylcyclohexane
Level: Developing Molecular Formula: C_7H_{14}

1. *Numbered Parent*



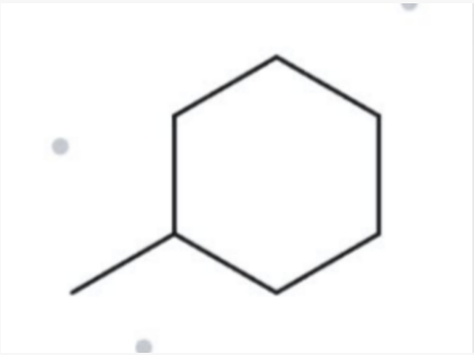
2. *Naming Trail*

- **Parent chain:** Cyclohexane (6 carbon atoms); the ring is the parent.
- **Numbering:** The carbon attached to the methyl group is C-1.
- **Substituent:** One methyl group at C-1. Final name: methylcyclohexane.

3. *My Name*

methylcyclohexane

4. *The Alkane Check*



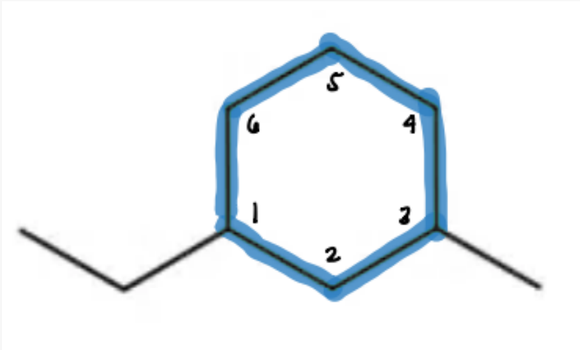
methylcyclohexane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A08 - 1-ethyl-3-methylcyclohexane
Level: Challenge Molecular Formula: C_9H_{18}

1. *Numbered Parent*



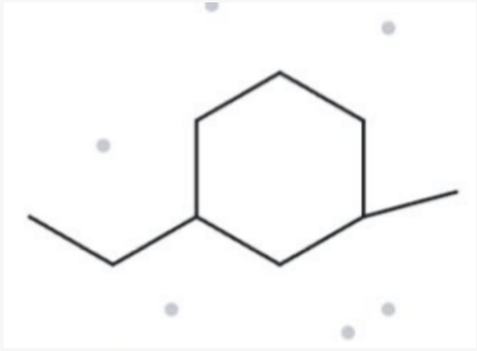
2. *Naming Trail*

- **Parent chain:** Cyclohexane (6 carbon atoms); the ring is the parent.
- **Numbering:** Begin at the carbon bearing ethyl and proceed toward methyl to obtain locants 1 and 3.
- **Substituents:** Ethyl at C-1 and methyl at C-3; alphabetical order is followed. Final name: 1-ethyl-3-methylcyclohexane.

3. *My Name*

1-ethyl-3-methylcyclohexane

4. *The Alkane Check*



1-ethyl-3-methylcyclohexane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A09 - 4-Isopropylheptane
Level: Challenge Molecular Formula: $C_{10}H_{22}$

1. *Numbered Parent*



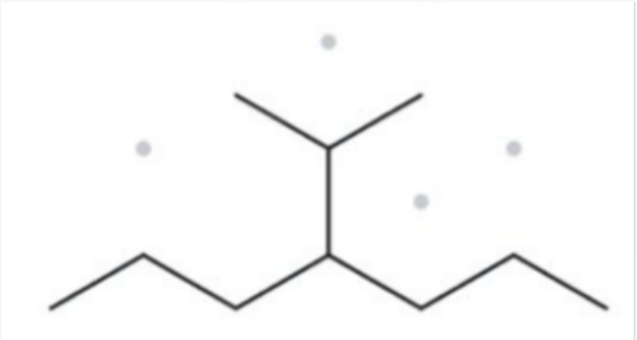
2. *Naming Trail*

- **Parent chain:** Heptane (7 carbon atoms).
- **Numbering:** Number from either end of the chain; the isopropyl substituent is located at C-4.
- **Substituent:** One isopropyl group at C-4. Final name: 4-isopropylheptane.

3. *My Name*

4-Isopropylheptane

4. *The Alkane Check*



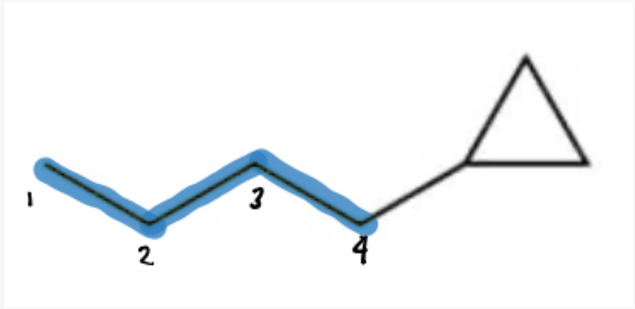
2-methyl-3-propylhexane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A10 - cyclopropylbutane
Level: Challenge Molecular Formula: C_7H_{14}

1. *Numbered Parent*



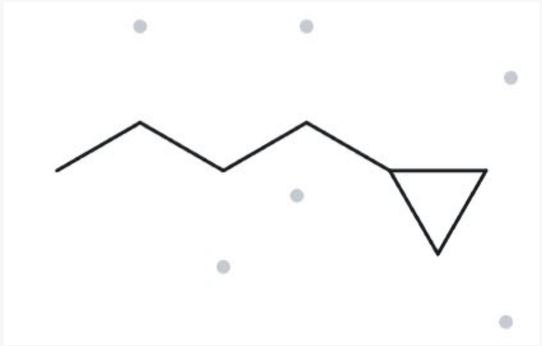
2. *Naming Trail*

- **Parent chain/ring:** Butane (4 carbon atoms) and cyclopropyl ring (3 carbon atoms).
- **Numbering:** The butane chain is numbered from the end attached to the cyclopropyl group, giving locant 1.
- **Substituent:** One cyclopropyl group at C-1. Final name: cyclopropylbutane.

3. *My Name*

cyclopropylbutane

4. *The Alkane Check*



butylcyclopropane

Verification Strip:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

NAME IT, DRAW IT!: ACTIVITY 3

PART B

NAME

FLIGHT

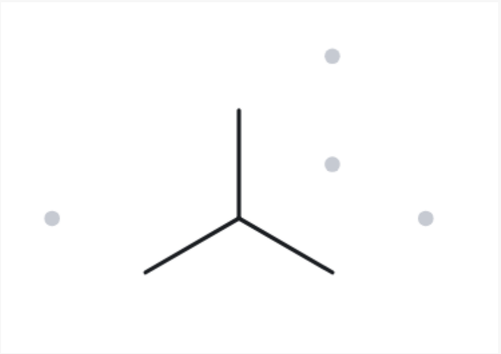
GATE

SEAT

CARD B01 - Given Name: 2-methylpropane
Level: Foundation

1. *Name Decoded*

- **Parent chain:** Propane, consisting of 3 carbon atoms.
- **Substituent:** One methyl group attached to carbon 2.
- **Expected molecular formula:** 3 parent carbons + 1 substituent carbon = 4 carbon atoms. Since it is an open-chain alkane, its formula is C_4H_{10}.



2. *Bond-line Structure*

3. *Formula Check*

- Formula obtained by counting the drawn structure: C_4H_{10}.
- Applying the open-chain alkane formula, C_nH_{2n+2}, gives C_4H_{10}.
- **Result:** The calculated formula agrees with the given compound.

4. *The Back Check*

- **Name generated by The Alkane: 2-methylpropane.**

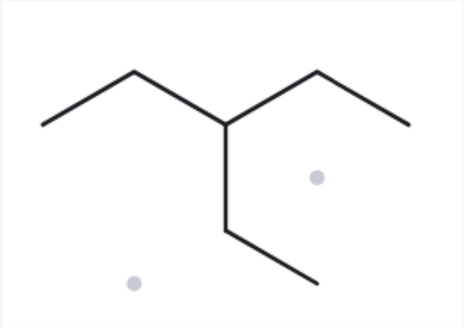
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B02 - Given Name: 3-ethylpentane
Level: Foundation

1. *Name Decoded*

- **Parent chain:** Pentane, containing 5 carbon atoms.
- **Substituent:** One ethyl group attached to carbon 3.
- **Expected molecular formula:** 5 parent carbons + 2 substituent carbons = 7 carbon atoms. The open-chain formula gives C_7H_{16}.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the structure:** C_7H_{16}.
- **Using C_nH_{2n+2} for an open-chain alkane gives C_7H_{16}.**
- **Result:** The calculated formula is consistent with the given name.

4. *The Back Check*

- **Name generated by The Alkane: 3-ethylpentane.**

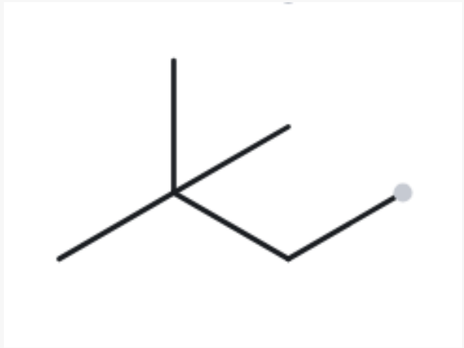
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B03 - Given Name: 2,2-dimethylbutane
Level: Foundation

1. *Name Decoded*

- **Parent chain:** Butane, consisting of 4 carbon atoms.
- **Substituents:** Two methyl groups attached to carbon 2.
- **Expected molecular formula:** 4 parent carbons + 2 substituent carbons = 6 carbon atoms. For an open-chain alkane, the formula is C_6H_{14}.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the structure:** C_6H_{14}.
- **Applying the general formula C_nH_{2n+2} gives C_6H_{14}.**
- **Result:** The calculated formula matches the given compound.

4. *The Back Check*

- **Name generated by The Alkane: 2,2-dimethylbutane.**

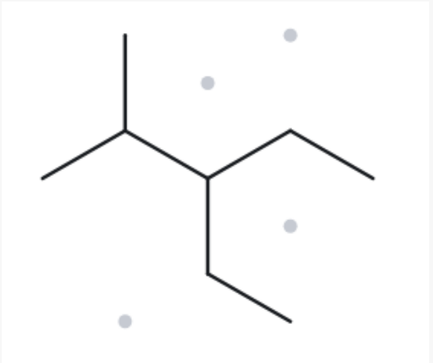
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B04 - Given Name: 3-ethyl-2-methylpentane
Level: Developing

1. *Name Decoded*

- **Parent chain:** Pentane, with 5 carbon atoms.
- **Substituents:** One methyl group at carbon 2 and one ethyl group at carbon 3.
- **Expected molecular formula:** 5 parent carbons + 2 ethyl carbons + 1 methyl carbon = 8 carbon atoms. Its open-chain formula is C_8H_{18}.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the drawn structure:** C_8H_{18}.
- **Using the general open-chain alkane formula gives** C_8H_{18}.
- **Result:** The calculated formula agrees with the given name.

4. *The Back Check*

- **Name generated by The Alkane: 3-ethyl-2-methylpentane.**

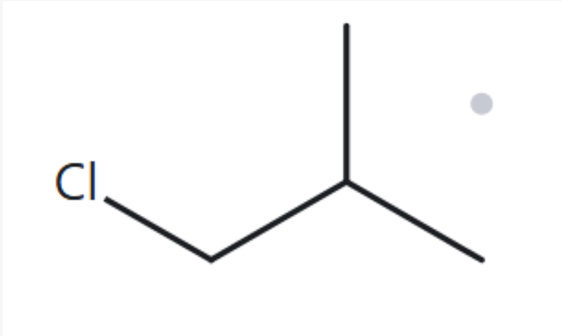
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B05 - Given Name: 1-chloro-2-methylpropane
Level: Developing

1. *Name Decoded*

- **Parent chain:** Propane, containing 3 carbon atoms.
- **Substituents:** One methyl group and one chlorine substituent.
- **Expected molecular formula:** 3 parent carbons + 1 methyl carbon = 4 carbon atoms. The corresponding alkane formula is C_4H_{10}. Replacing one hydrogen atom with chlorine produces C_4H_9Cl.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the structure:** C_4H_9Cl.
- **The open-chain alkane formula gives C_4H_{10}. Replacing one hydrogen with chlorine results in C_4H_9Cl.**
- **Result:** The calculated formula matches the given compound.

4. *The Back Check*

- **Name generated by The Alkane: 1-chloro-2-methylpropane.**

Verification Strip:

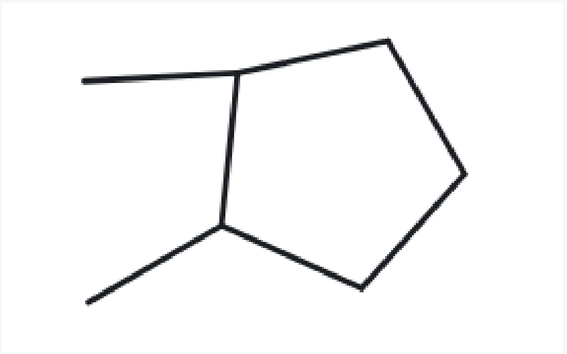
✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B06 - Given Name: 1,2-dimethylcyclopentane
Level: Developing

1. *Name Decoded*

- **Parent ring:** Cyclopentane, containing 5 carbon atoms.
- **Substituents:** Two methyl groups attached at carbons 1 and 2.
- **Expected molecular formula:** 5 ring carbons + 2 substituent carbons = 7 carbon atoms. Since the compound contains one ring, its formula is C_7H_{14}.

2. *Bond-line Structure*



3. *Formula Check*

- **Formula counted from the drawn structure:** C_7H_{14}.
- **Applying the one-ring hydrocarbon formula, C_nH_{2n}, gives C_7H_{14}.**
- **Result:** The calculated formula is consistent with the given name.

4. *The Back Check*

- **Name generated by The Alkane: 1,2-dimethylcyclopentane.**

Verification Strip:

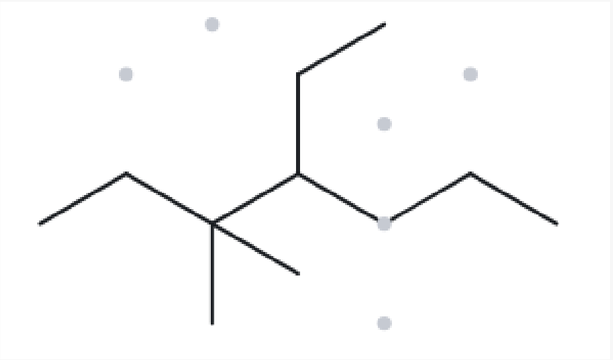
✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B07 - Given Name: 4-ethyl-3,3-dimethylheptane
Level: Developing

1. *Name Decoded*

- **Parent chain:** Heptane, consisting of 7 carbon atoms.
- **Substituents:** Two methyl groups at carbon 3 and one ethyl group at carbon 4.
- **Expected molecular formula:** 7 parent carbons + 2 ethyl carbons + 2 methyl carbons = 11 carbon atoms. The open-chain formula gives $C_{11}H_{24}$.

2. *Bond-line Structure*



3. *Formula Check*

- **Formula counted from the structure:** $C_{11}H_{24}$.
- **Using C_nH_{2n+2} for an open-chain alkane gives $C_{11}H_{24}$.**
- **Result:** The calculated formula agrees with the given name.
-

4. *The Back Check*

- **Name generated by The Alkane: 4-ethyl-3,3-dimethylheptane.**

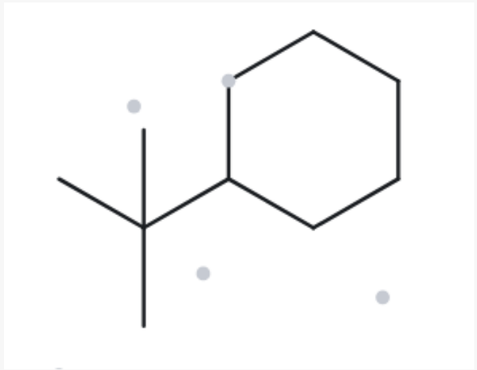
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B08 - Given Name: tert-butylcyclohexane
Level: Challenge

1. *Name Decoded*

- **Parent ring:** Cyclohexane, containing 6 carbon atoms.
- **Substituent:** One tert-butyl group attached to carbon 1.
- Expected molecular formula: 6 ring carbons + 4 substituent carbons = 10 carbon atoms. Since the compound has one ring, its formula is $C_{10}H_{20}$.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the structure: $C_{10}H_{20}$.**
- **Applying the one-ring hydrocarbon formula gives $C_{10}H_{20}$.**
- **Result:** The molecular formula agrees with the given compound, although the website presents a different systematic name.

4. *The Back Check*

- **Name generated by The Alkane: (1,1-dimethylethyl)cyclohexane.**
- **The returned name uses a systematic alternative for the tert-butyl substituent.**

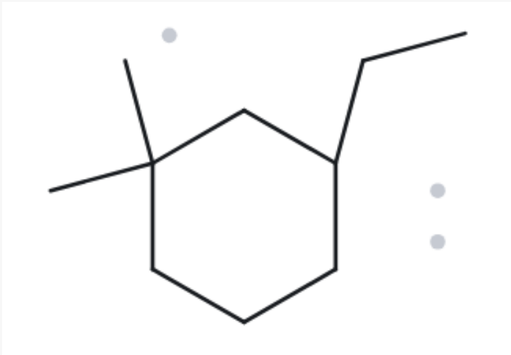
Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B09 - Given Name: 3-ethyl-1,1-dimethylcyclohexane
Level: Challenge

1. *Name Decoded*

- **Parent ring:** Cyclohexane, containing 6 carbon atoms.
- **Substituents:** Two methyl groups at carbon 1 and one ethyl group at carbon 3.
- **Expected molecular formula:** 6 ring carbons + 2 methyl carbons + 2 ethyl carbons = 10 carbon atoms. Its one-ring formula is $C_{10}H_{20}$.



2. *Bond-line Structure*

3. *Formula Check*

- **Formula counted from the drawn structure:** $C_{10}H_{20}$.
- **Using the one-ring hydrocarbon formula, C_nH_{2n}, gives $C_{10}H_{20}$.**
- **Result:** The formula and returned name agree with the given compound.

4. *The Back Check*

- **Name generated by The Alkane: 3-ethyl-1,1-dimethylcyclohexane.**

Verification Strip:

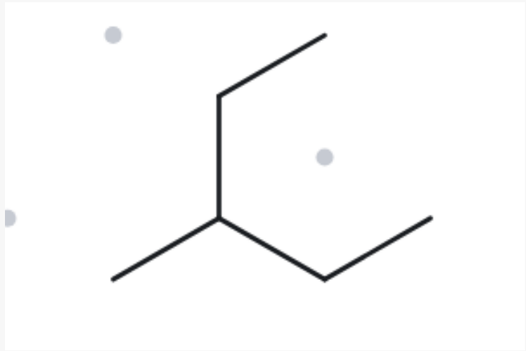
✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED

CARD B01 - Given Name: 2-ethylbutane
Level: Challenge

1. *Name Decoded*

- **Longest parent chain:** Pentane, containing 5 carbon atoms.
- **Substituent:** One methyl group at carbon 3.
- **Expected molecular formula:** The original name represents 4 butane-chain carbons + 2 ethyl-group carbons, giving 6 carbon atoms in total. The resulting open-chain formula is C_6H_{14}.

2. *Bond-line Structure*



3. *Formula Check*

- **Formula counted from the drawn structure:** C_6H_{14}.
- **Applying the open-chain alkane formula gives** C_6H_{14}.
- **Result:** The molecular formula agrees, but the supplied name is deliberately incorrect. The corrected name is 3-methylpentane.

4. *The Back Check*

- **Name generated by The Alkane: 3-methylpentane.**
- **The returned name differs from the supplied name because the longest continuous chain must be selected when naming the structure.**

Verification Strip:

✓ PARENT LENGTH MATCHES | ✓ SUBSTITUENT POSITIONS MATCH | ✓ FORMULA AGREES | ✓ BACK-CHECK RECORDED